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ORIGINAL RESEARCH

QViLa: Quantum Infused Vision-Language Model for Enhanced Multimodal Understanding

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Abstract

Vision-language models have emerged as transformative tools, revolutionizing the integration of visual and textual information, forging pathways for nuanced interpretations across various applications. The evolution of these models underscores the challenge of achieving seamless modality fusion, particularly in aligning raw pixel values of images with high-level semantics of text, a bottleneck that often hinders optimal cross-modal representations. Addressing this, our research introduces a quantum-enhanced multimodal framework. The main objective of our proposed work is the integration of classical vision-language transformers with quantum-augmented layer, aimed at enhancing the fusion of extracted feature embeddings, thereby bridging the modality gap. The quantum computing techniques, offers an innovative approach to information processing, which paves the way for richer and more intricate visual-textual representations. Furthermore, the shared self-attention mechanism accentuates the model's ability to detect complex modality interactions. The quantum-enhanced framework is empirically evaluated on the VQA v2 dataset. This evaluation not only considers accuracy across diverse question categories but also the model's computational efficiency, emphasizing the pivotal contributions of quantum computations in achieving heightened accuracy levels. Further exploration into the influence of different quantum feature maps aided in identifying the most optimal model variant. Our findings highlight the quantum layer's pivotal role in improving the efficacy of classical vision language models.

Keywords Multimodal learning · Quantum computing · Vision language model · Visual reasoning

Introduction

In recent years, quantum-based models have risen to prominence [1–4], driven by the increased availability of quantum hardware and the growing expertise in quantum algorithms. These models leverage the unique properties of quantum mechanics, such as superposition and entanglement, to process information in fundamentally distinct ways compared to their classical counterparts. Quantum-based neural networks have garnered significant attention as a promising approach to solving computer vision tasks [5–7]. This paradigm shift from classical to quantum computing has the potential to revolutionize the way vision-language (VL) models learn cross-modal representations.

VL models have been gaining significant attention for aiding in many pivotal applications [8–10]. Multimodal models, which fall under this category, harness the capability of integrating information from multiple sources, often comprising text and images, to generate richer and more coherent representations. These models aim to bridge the semantic gap between visual and linguistic modalities by jointly embedding images and their corresponding text. Historically, the challenge lies in aligning the disparate nature of raw pixel values in images with the high-level abstract semantics of text. With the emergence of advanced deep learning architectures such as vision transformers for images [11] and language transformers like BERT (Bidirectional Encoder Representations from Transformers) for text

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[12–14], the integration between visual and textual information has been progressively enhanced, enabling models to achieve superior results across various VL tasks.

Quantum-enhanced models offer a novel perspective on how information processing can be augmented. While applying quantum computing to neural networks and machine learning is still in its nascent stages, preliminary studies suggest that these models can offer exponential speedups and enhanced capacity for specific tasks [15, 16]. By entangling quantum states, it becomes possible to encode and process vast amounts of information simultaneously, thereby allowing for richer and more convoluted representations, especially in high-dimensional spaces like visual modality. In the context of multimodal models, integrating quantum-enhanced techniques for visual representation can pave the way for revolutionary improvements in accuracy and efficiency, setting the stage for a new generation of VL models.

Hybrid neural networks, which integrate classical neural networks with quantum layers, represent a notable advancement in vision-language tasks. These models leverage the strengths of both classical deep learning and quantum computing, enabling them to address the intrinsic complexities of multimodal data more effectively. Recent studies [17, 18] have shown that hybrid models can significantly enhance the expressiveness and generalization capabilities of vision-language models by incorporating quantum features that capture intricate patterns and dependencies. The benefits of hybrid quantum-classical neural networks lie in their ability to process information in parallel through quantum superposition and entanglement, thereby providing a deeper and more nuanced understanding of cross-modal interactions. This approach not only improves computational efficiency but also paves the way for innovative applications that require high-dimensional data processing, such as visual question answering and image captioning.

In this study, we introduce a quantum-enhanced multimodal framework. By leveraging the strengths of the vision transformer model for visual interpretation and a state-of-the-art language transformer for textual nuances, we address challenges faced in contemporary modality fusion techniques. The integration of the feature embeddings from these transformers is subsequently processed through a dedicated quantum layer. Given the inherent complexity in fusing these modalities, the quantum intervention offers unparalleled depth in aligning visual and linguistic embeddings. To further refine and align the joint representation between visual and linguistic information, a multi-head self-attention mechanism is incorporated. This strategy is pivotal for capturing complex modality interactions and nuances, which are often overlooked in conventional models. We evaluate our model on the Visual Question Answering (VQA) v2 dataset [19], comparing its performance with other state-of-the-art models. The findings provide significant insights into the potential of quantum-enhanced VL models, potentially steering future research directions in this domain.

Related Works

In the domain of VL pretraining, there has been a notable evolution from earlier approaches reliant on region-based feature extraction through object detectors [20, 21] to the contemporary prevalence of transformer-based models. Concurrently, innovative techniques have emerged, such as context optimization, which focuses on learning continuous vectors for task-specific prompts, avoiding the complexities of prompt engineering. An alternative approach, CLIP-Adapter [22], conducts fine-tuning using feature adapters on either the visual or language branch, demonstrating superior performance and simplicity compared to context optimization. CLIP's alignment of images and texts in a shared feature space presents challenges with prompt engineering, where minor wording changes can drastically alter performance. Addressing this, Zhou et al. [23] introduced Context Optimization (CoOp), an adaptation for CLIP-like models that models prompt context with learnable vectors, offering significant performance boosts over traditional prompt engineering. Test-time prompt tuning [24], further advances by learning adaptive prompts from a singular test sample and enhances CLIP's zero-shot top-1 accuracy by an average of 3.6%, outperforming methods that depend on additional task-specific training data.

Advancements in VL pretraining emphasize the significance of visual features and their representations in VL models. The improvement of visual representations specifically for VL tasks has been shown to yield richer visual object detection when paired with transformer-based VL fusion models [21]. While large-scale VL pre-training methods have conventionally relied on the exact correspondence between image-text pairs, PyramidCLIP [25] employs a hierarchical alignment technique for the visual and linguistic elements, surpassing the performance of CLIP in ImageNet zero-shot classification. Furthermore, Flamingo [26] enables rapid adaptation to various image and video tasks using few-shot learning, excelling in tasks such as visual reasoning and captioning with minimal task-specific data. In contrast to the trend of large-scale pretraining [27], proposes BLIP-2, a pretraining strategy that efficiently bootstraps VL pre-training using off-the-shelf frozen models, showcasing superior performance on various tasks with significantly fewer trainable parameters than existing methods.

VL models have also been extensively utilized in the medical field, particularly in medical VQA [28]. While

general-domain VQA is well-explored, medical VQA poses unique challenges, necessitating specialized datasets and approaches. Numerous studies have explored diverse approaches, from leveraging multi-level visual features using the Feature Pyramid Network [29] to adopting a transformer encoder-decoder architecture that combines the strengths of vision transformers for image features and textual encoder transformers for questions [30]. Furthermore, the introduction of the Patientoriented Visual Question Answering (P-VQA) dataset [31] represents a significant stride, covering an entire treatment process and encompassing a medical knowledge graph. Researchers have also identified medical-specific challenges and discussed future research directions, aiming to motivate further investigation in the field of medical VQA. These advances highlight the significance of domain-specific methodologies and the potential of comprehensive datasets in enhancing medical VQA systems.

In disaster management, the integration of VL models has shown promise in improving response and recovery efforts [32]. For instance, addressing the challenges in VQA on remote sensing (RSVQA), efforts have been made to enhance visual-spatial reasoning capabilities [33, 34]. Techniques like supervised attention-based VQA (SAM-VQA) [35] have been employed, leveraging drone imagery for effective post-disaster damage assessment. These models surpass traditional attention-based methods by deriving visual attention based on questions, thereby providing more accurate and trustworthy answers for decision-making. Furthermore, there's a growing emphasis on the robustness of RSVQA models. The need to learn robust representations against new words and varying question templates has led to the proposal of contrastive learning strategies [36]. These strategies, along with augmented datasets, demonstrate improved performance and robustness, especially on low-resolution datasets, enhancing the reliability of RSVQA models. These diverse approaches highlight the ongoing advancements and the potential of visual language models in specialized and critical applications.

The integration of visual and textual data plays a crucial role in VL models, where attention mechanisms and fusion strategies have emerged as pivotal elements. Recent developments have highlighted the growing potential of attention mechanisms in enhancing cross-modal retrieval, leading to smarter and more human-like VQA systems [37, 38]. Transformers [39, 40], renowned for their self-attention capabilities, have been adapted in various VQA models, with innovations such as the Local SelfAttention in Transformer (LSAT) [41] addressing the balance between global and local dependency modeling. This approach allows for effective capturing of contextual information while avoiding redundancy. Performance improvements have been achieved in VQA by concentrating attention through threshold-based methods, effectively filtering out irrelevant image and question features, thus refining the focus of the model [42]. Addressing the complexities of event-level VQA, some models have introduced frameworks for cross-modal causal relational reasoning [43]. This involves causality-aware reasoning and spatial-temporal transformers, enhancing the understanding of visual-linguistic causal structures and improving robustness. Another significant advancement is the incorporation of scene text features in VQA models, addressing the previous limitations in text reading abilities. By integrating Optical Character Recognition (OCR) tokens through co-attention mechanisms [44], these models have shown improved performance, demonstrating the ability to answer questions related to text in images more accurately. Moreover, these advancements indicate a trend towards more refined and contextually aware VQA systems, with attention mechanisms and fusion strategies playing a crucial role in enhancing the multimodal understanding and interpretability of these models.

Quantum models have been extensively explored in various fields, with recent advancements focusing on the development of quantum neural networks (QNNs) inspired by classical neural architectures to address complex problems in quantum physics and beyond. A notable work in this domain is the Quantum Convolutional Neural Network (QCNN) [45], which demonstrates superior capabilities in recognizing quantum states and optimizing quantum error correction schemes. Another study [46] benchmarks fully parameterized QCNNs for classical data classification, demonstrating excellent accuracy on MNIST and Fashion MNIST datasets, with shallow-depth quantum circuits. Additionally, Quantum Language Models (QLMs) [47] with an entanglement embedding module have been proposed to transform word sequences into entangled pure state representations. These models effectively capture non-classical correlations and have shown superior performance on question answering tasks. Quantum machine learning (QML) involves hybrid quantum-classical frameworks, where variational quantum circuits are trained classically and quantum components are integrated for enhanced performance. One such technique is classical-quantum transfer learning, which uses pre-trained classical models combined with quantum circuits at the decision layer for high-precision classification. This approach has been successfully applied to Natural Language Processing (NLP), utilizing pre-trained BERT models and fine-tuned variational quantum circuits to classify short texts [48]. Wang et al. [49] propose quantum adversarial transfer learning, where adversarial training processes are adapted to quantum states, demonstrating exponential advantages in computing resources and achieving high accuracy in various datasets through quantum-encoded

data and measurement-based judgments. Furthermore, the classical-quantum transfer learning paradigm has shown significant improvements in computer vision tasks and is now being extended to NLP. By pre-training sentence states with complex-valued BERT-like architectures and adapting them to quantum transfer learning, substantial performance increases have been observed in quantum simulation experiments [50], highlighting the potential of hybrid models in enhancing NLP applications.

There have been efforts to merge quantum mechanics with classical transformer architectures, taking advantage of the high performance these offer for tasks like image analysis and NLP [51]. Moreover, the classical counterparts have witnessed quantum adaptations, where shallow quantum circuits are employed to provide different classification models, potentially with fewer parameters than classical versions. Quantum graph transformers [52] have demonstrated their efficacy in tasks like graph classification, emphasizing the benefits of quantum integration. As quantum computing intersects with NLP, innovative hybrid strategies have been developed to process largescale data [3, 53]. The hybrid workflows introduced not only allow for the encoding, and processing of textual data using quantum circuit models, but also demonstrate effectiveness in tasks such as understanding and comparison of sentence structures. Furthermore, the rise of quantum machine learning demonstrates the rapid computational advantages, particularly evident in image processing tasks [54]. Quantum image processing offers enhanced techniques for image encryption and segmentation, due to the parallel computing capabilities of quantum systems. In parallel, the integration of quantum principles into neural networks has been pursued, with new models emphasizing reduced memory requirements and faster optimization [55]. These developments highlight the transformative potential of quantum models in contemporary computing, enhancing the capabilities of transformers and neural networks.

Methodology

Quantum Computational Framework in VL Models

The limitations inherent to classical computing paradigms demands exploration into alternative computational methodologies. Quantum mechanics, with its foundational concept of the quantum Hilbert space, serves this purpose by encapsulating the expansive state space of quantum systems. The classical data is mapped into the quantum Hilbert space via a process termed as quantum encoding. Following this encoding, the data is manipulated using specialized quantum gates and circuits. Quantum gates are the building blocks of quantum circuits, analogous to classical logic gates in traditional computing. These gates manipulate quantum bits (qubits) through unitary transformations, preserving the quantum state's probabilistic nature. This allows for the leveraging of the quantum phenomena, superposition and entanglement. In superposition, qubits have the capability to concurrently represent multiple states. Unlike classical bits, which are either 0 or 1, qubits can be in a superposition of both states. Entanglement is the state of mutual dependence between qubits irrespective of the spatial distance between them, such that the state of one qubit directly affects the state of the other. This quantum computational framework lays the groundwork for the enhanced capabilities of our proposed VL models.

Quantum Feature Maps

Feature maps act as essential connectors, translating classical data representations into quantum states. Specifically, complex quantum feature maps provide an advanced encoding mechanism, making them crucial for tasks requiring detailed data mappings. In our quantum-enhanced VL model, these feature maps play a crucial role. By harnessing their capabilities, we adeptly convert classical features extracted from various modalities into quantum states. This conversion enhances the multidimensionality of the original data, allowing for richer and more detailed comparisons in the quantum domain. Therefore, in the blend of classical and quantum models, complex quantum feature maps stand as a key component, ensuring that data from diverse sources can be efficiently integrated, and processed in our multimodal quantum framework.

Z-Feature Map

The Z-Feature map (ZFM) provides a novel approach to map classical data, especially from image sources. This is crucial in a setting where classical and quantum models are used concurrently to harness both classical computing's efficiency and quantum computing's parallelism. This entangled mapping serves as an essential bridge, ensuring that data extracted from various modalities are coherently and efficiently integrated within our quantum framework.

By embedding the classical image vectors into quantum states through the ZFM, we are better positioned to compare and integrate data across modalities and models.

The Pauli-Z gate serves as the foundation for the encoding process in ZFM, ensuring a nuanced transformation of the classical data.

The unitary operator for the Z-feature map, denoted as $U^{Z}_{\phi(\vec{F})}$ is represented by Eq. (1)

$$U^{Z}_{\phi(\vec{F})} = \exp\left(i\sum_{j}\phi_j(\vec{F})Z_j\right)H^{\otimes n} \tag{1}$$

where $\phi_j((\vec{F}))$ is the function representing the *j*-th feature of the vector $(\vec{F})$, and $H^{\otimes n}$ represents the tensor product of the Hadamard gate H applied across n qubits. The feature mapping operation ϕ, converts the classical feature vector into parameters suitable for quantum operations.

ZZ-Feature Map

The ZZ Feature Map (ZZFM) offers another pivotal bridge between classical and quantum data. The basis for this map hinge on the interactions between the pairs of qubits, leveraging the tensor product of two Pauli-Z gates. This interconnections makes it suitable for tasks that necessitate understanding relations within data, which is crucial in multimodal setups.

The unitary transformation for the ZZFM is mathematically represented as Eq. (2)

$$U^{ZZ}_{\phi(\vec{F})} = \exp\left(i\sum_{j,k}\phi_{jk}(\vec{F})Z_j\otimes Z_k\right) \tag{2}$$

where $Z_j \otimes Z_k$ represents the tensor product of the Pauli-Z operator on qubits *j* and *k*. The tensor product indicates that the operation is applied to multiple qubits simultaneously.

Pauli-Feature Map

The Pauli Feature Map represents a comprehensive quantum encoding mechanism, taking cues from the entire Pauli group. This group comprises the Pauli-X, PauliY, and Pauli-Z matrices, thus ensuring an expansive representation. By utilizing the full breadth of the Pauli group, this feature map offers a richer encoding spectrum, potentially capturing intricate nuances and details inherent in multimodal datasets.

Mathematically, the PFM is expressed as Eq. (3)

$$U^{Pauli}_{\phi(\vec{F})} = \exp^{(i\sum_j[\phi_{jx}(\vec{F})X_j+\phi_{jy}(\vec{F})Y_j+\phi_{jz}(\vec{F})Z_j])} \tag{3}$$

These feature maps offer a systematic approach for encoding classical data into the quantum domain. The representations are specifically designed to address the intricacies inherent in merging different modalities. The selection between these feature maps is guided by the data's characteristics and the desired intricacy of the quantumclassical interplay being investigated.

Shared Self Attention for Multimodal Fusion

In the proposed architecture, an essential component that facilitates the fusion of vision and language modalities is the shared self-attention mechanism. This mechanism is designed to capture interdependencies between the image and text embeddings, thus enhancing the combined representation's richness and depth. It allows the model to weigh the importance of different parts of an input sequence relative to each other. In our multimodal scenario, shared self-attention amplifies this principle by entangling both visual and textual information. This entanglement ensures that the fused representation is not just a simple concatenation operation, but is contextually enriched with information from both modalities. Mathematically, shared self-attention is represented by Eq. (4), where Q, K, and V denote queries, keys, and values.

$$Attention\,(Q,\,K,\,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

where, d_k is the dimensionality of the key vectors, and it acts as a scaling factor to ensure that the dot products used to compute the attention scores do not grow too large. Incorporating shared self-attention into our model assures that the relationships and semantics across modalities are well understood and represented.

Proposed Methodology

In the proposed framework, a multimodal approach is adopted to seamlessly integrate visual and textual information sources as illustrated in Fig. 1. The multimodal input encompassing image and its associated text is given to the vision and language transformer for processing. These transformers work on their respective modalities to process the input and produce corresponding embedding representations. The resulting embeddings capture the salient features and patterns inherent within the visual and linguistic data. Upon extraction of these embeddings, they are concatenated, creating a single integrated representation that encompasses information from both the image and the query text.

The concatenated representation is channeled into a 4 qubit quantum layer as represented in Fig. 2. Within the quantum layer, the angle embedding mechanism encodes the input features by applying rotations using specific quantum gates. This encodes classical data into the amplitudes of a quantum state. The input features are transformed into a quantum-compatible format that can be effectively processed within the quantum circuit. Following this embedding phase, the entangler layers establish entanglements between the quantum states, which establishes correlations

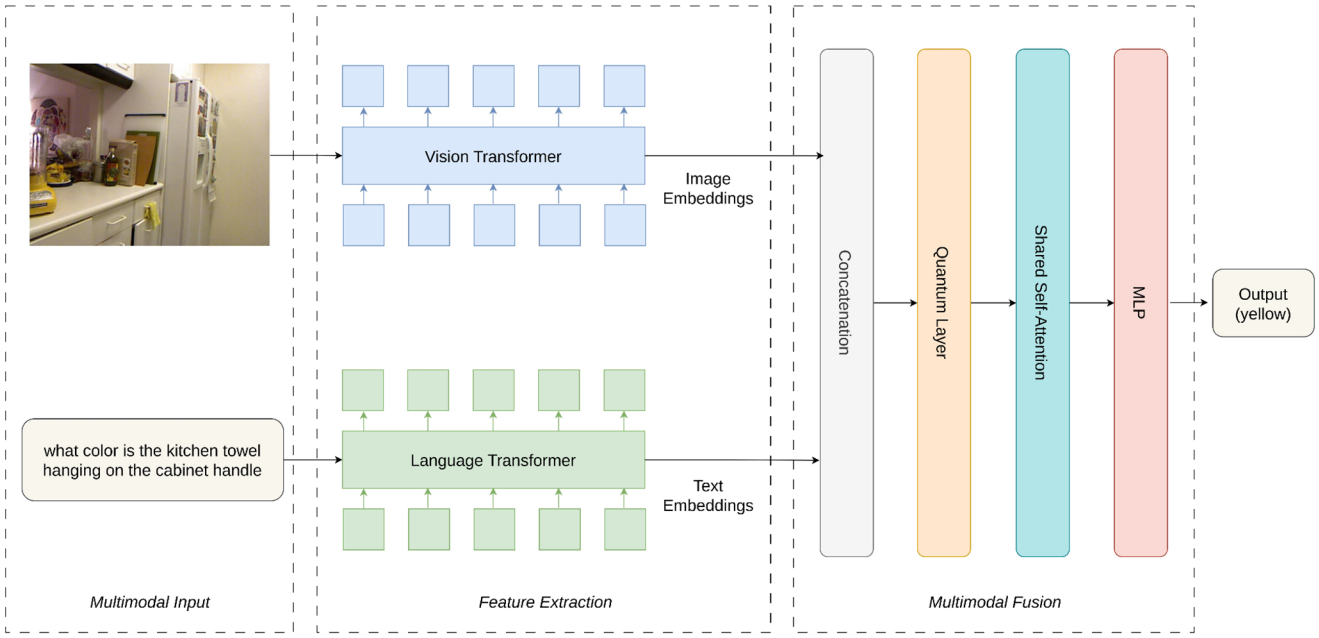


Fig. 1 Architecture of the proposed multimodal integration framework

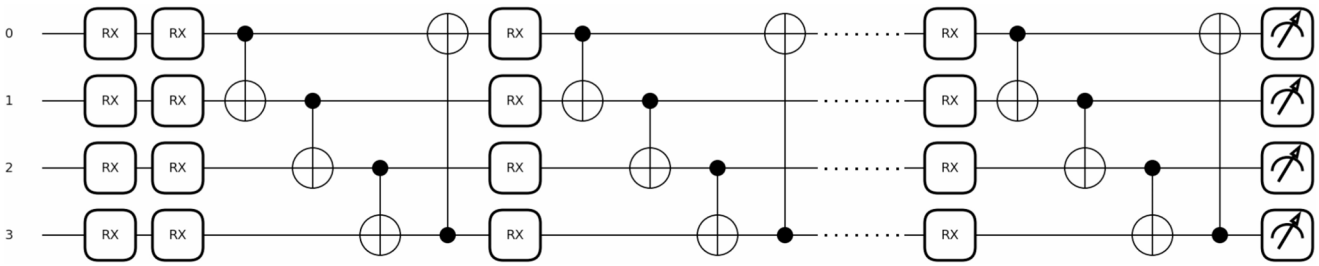


Fig. 2 4-qubit quantum circuit layer with angle embedding and entangling gates

amongst them. This entanglement not only emphasizes the richness of representation but also amplifies the depth of feature interaction. Following this phase, quantum feature maps are applied to further refine the quantum state. These feature maps reorient and transform the quantum data. Finally, the quantum layer returns the computation of expectation values, from each qubit, presenting a set of outputs representing the quantum-processed embeddings. The model processes visual and textual data through their respective encoders, producing a 768-dimensional feature vector from the vision encoder and a 256-dimensional feature vector from the language encoder. The concatenation of these vectors results in a fused feature vector of 1024 dimensions. Given the limitations of current quantum hardware, directly encoding a 1024-dimensional feature vector into a quantum circuit would necessitate an impractically large number of qubits. To address this issue, we reduce the dimensionality of the fused feature vector using a fully connected layer, compressing the 1024-dimensional vector to a 4-dimensional vector that can be feasibly input into the 4-qubit quantum layer.

Post quantum processing, the data is given to the shared self attention module. The attention mechanism provides weightage to the quantum embeddings, focusing on the most relevant quantum features in relation to the VQA task. This ensures that the most relevant quantum-derived patterns are amplified while less relevant features are attenuated. The attention-refined quantum embeddings traverse into the Multi-Layer Perceptron (MLP) layer. This final layer, embedded with classification capabilities, interprets the quantum-enhanced embeddings to render a decisive judgment pertaining to the VQA task, thereby identifying the most suitable response to the query posed against the provided image. The MLP layer in the model pipeline is a crucial component for the final classification in the VQA model. After the concatenated 1024-dimensional feature vector is processed through the 4-qubit quantum layer, the quantum-processed features are transformed by computing the expectation values of the quantum states. These

processed features are then integrated into the shared self-attention mechanism, which enhances the interaction and dependencies among the features. The refined features from the attention mechanism are subsequently passed into the MLP for final classification. The MLP architecture consists of a first fully connected layer that maps the input to 512 dimensions, followed by a ReLU activation function and a dropout layer with a probability of 0.5 to prevent overfitting. This is followed by a second fully connected layer that reduces the 512-dimensional vector to 256 dimensions, again followed by a ReLU activation function. The final output layer maps the 256-dimensional vector to 3129 dimensions, corresponding to the number of answer choices in the VQA task. The MLP is trained using the Adam optimizer with a learning rate of 1e-4. The model training process spans 50 epochs, and the cross-entropy loss function is used for multi-class classification.

Results & Discussion

This section presents the empirical assessment of our proposed model, benchmarking its performance using evaluation metrics such as accuracy and computational speed. For classical computations, the experiments were conducted on NVIDIA Tesla P100 GPU equipped with 16 GB of memory. Meanwhile, quantum computations were executed on the Aer IBM state vector simulator using a 4-qubit quantum processor configuration.

Experimental Setup

In our experimental analysis, we evaluated the performance of our proposed multimodal model across different configurations, involving the interchange of vision and language transformers. The model underwent training on the VQA v2 dataset, and we assessed its accuracy across a spectrum of question categories, encompassing yes/no queries, numerical inquiries, and others. Furthermore, we compared the training duration for each of these model variations, to examine the computational efficiency of each configuration.

Table 1 Comparison of different model variants on VQA v2 test-std

Model	Yes/No	Number	Other	Overall
DeiT + BERT	87.48	55.12	59.61	70.85
ViT + BERT	89.33	58.06	63.85	73.51
BEiT + BERT	89.05	58.39	63.21	73.19
DeiT + ALBERT	83.96	49.16	55.87	65.10
ViT + ALBERT	85.32	48.92	57.69	66.97
BEiT + ALBERT	85.70	51.49	59.13	68.78
DeiT + RoBERTa	90.14	58.61	65.21	74.32
ViT + RoBERTa	90.55	58.43	65.89	75.26
BEiT + RoBERTa	91.73	60.07	65.33	76.36

To further investigate the potential of quantum enhancements, we conducted a comprehensive examination of the impact of different feature maps on our model's performance. These experiments were conducted to evaluate various model variants and identify the superior model based on VQA task accuracy and computational performance. All the quantum simulations were conducted in a noiseless environment.

The transformer model variants were structured with 12 Transformer layers, each comprising 768 hidden units and 12 attention heads. Image inputs were standardized to a resolution of 224×224, and during the training phase, a patch size of 16×16 was utilized. Each tokenizer, corresponding to the different language model variants, was restricted to a maximum text sequence length of 25. We adopted whole word masking as a strategy for the masked language modeling pre-training task. The training process spanned 50 epochs with a batch size of 32. The models were optimized using the AdamW optimizer with a learning rate of 1e-4, and the loss was calculated using the cross-entropy loss function. The dataset comprised 82 K images, with a $60-40$ split designated for training and validation respectively. We approached the VQA challenge as a classification exercise, with each potential answer representing a class. The cfinal fused representations were given to MLP to predict the answer from a pool of 3,129 classes. The final decision-making during inference was based on selecting the answer exhibiting the highest confidence.

Comparative Analysis of Model Variants

We explored the combination of various vision transformer variants with different language transformers to examine their efficiency in VQA task. Table 1 presents the performance metrics of various model configurations evaluated on the VQA v2 test-std dataset. For this empirical assessment, we considered various variations of vision transformers including DeiT [56], ViT [11], and BEiT [39], combined with language transformers such as BERT [12], ALBERT [57], and RoBERTa [13]. We found that the combination of BEiT with RoBERTa emerged as the top-performing model variant, achieving an overall accuracy of 76.36%. When assessed category-wise, the model scored an accuracy of 91.73% for yes/no questions, 60.07% for number-based questions, and 65.33% for queries in other categories.

In contrast, the combination of DeiT and ALBERT yielded the lowest overall accuracy of 64.73%. The reduced performance can be attributed to the inherent characteristics of ALBERT. As a streamlined version of BERT, ALBERT is designed to provide similar capabilities but with fewer parameters. This reduction might have led to a compromise in the model's capacity to capture complex patterns

essential for the VQA task. The BERT model variants typically outperform ALBERT counterparts and are on par with RoBERTa variations. This suggests that the added depth and complexity of BERT provides a better complementary structure when fused with vision transformers.

Figure 3 illustrates the comparative training plots for Accuracy and Loss metrics of various model variants. Taking a closer look at the performance metrics, the BEiT variant, when paired with RoBERTa, achieved the highest accuracy of 76.36%. In contrast, the ViT model, also combined with RoBERTa, scored an overall accuracy of 75.23%, making it the next best-performing model. When combined with BERT variant, the ViT model managed to surpass the BEiT variant. This superiority of ViT in such a combination could be due to its generalized architecture, which might be better attuned to BERT's linguistic representations compared to BEiT's specialized image embeddings. Meanwhile, the DeiT variant, even when combined with high-performing language models, did not perform better. This is due to the model's primary design focus on data efficiency rather than intricate pattern recognition crucial for VQA tasks.

Analyzing the difference in performance between the top and bottom model variants, there's a notable increase of approximately 11.26% in overall accuracy. This emphasizes the necessity of selecting the appropriate model combination for tasks demanding a detailed comprehension of both visual and textual elements. BEiT's unique method of integrating visual embeddings aligns well with RoBERTa's

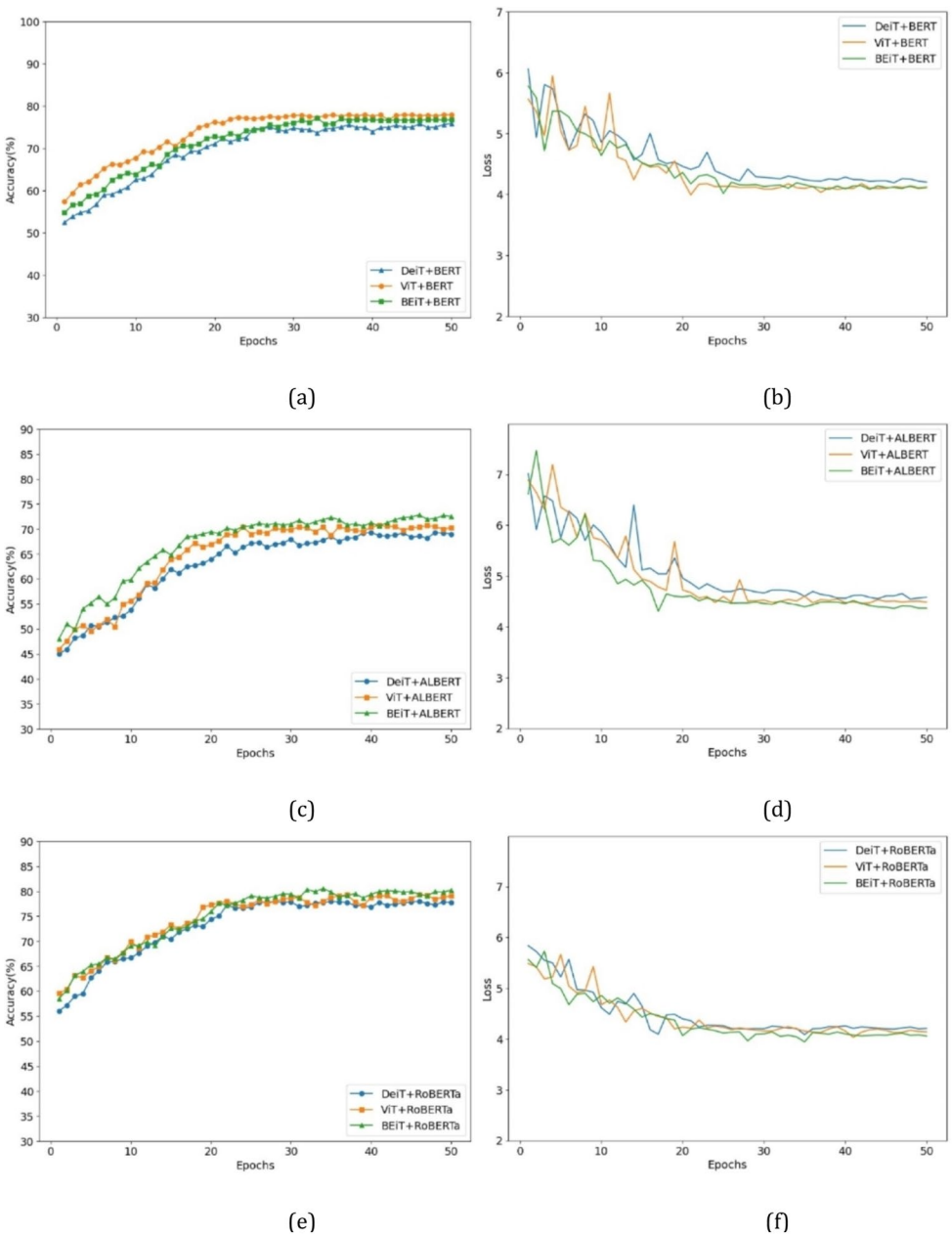


Fig. 3 Training performance over 50 epochs for (**a**) accuracy plot of BERT variant (**b**) loss plot of BERT variant (**c**) accuracy plot of ALBERT variant (**d**) loss plot of ALBERT variant (**e**) accuracy plot of RoBERTa variant (**f**) loss plot of RoBERTa variant

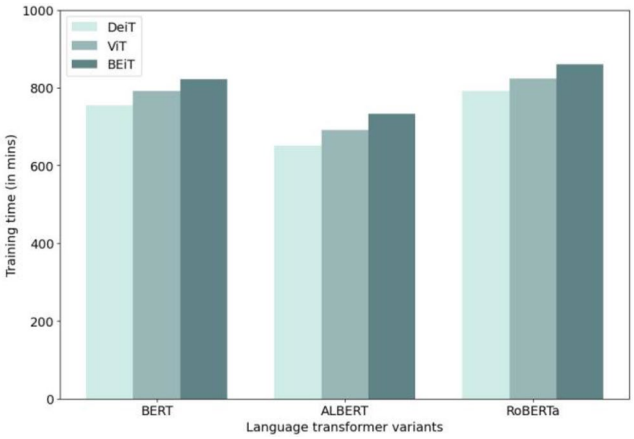


Fig. 4 Training duration across different model variants

detailed language modelling, leading to their combined proficiency. Overall, while all vision transformer variants are good at processing visual information efficiently, their individual architectures, combined with their interaction with different language models, significantly influence their performance in multimodal scenarios.

In the experimental results presented in Fig. 4, the training time for various model variants is depicted. For model configurations combined with BERT, the BEiT variant demanded the most extended training duration, clocking in at approximately 810 min. Conversely, the ViT and DeiT configurations exhibited a notably reduced training duration. When paired with ALBERT, the training times for BEiT and ViT were relatively close, highlighting their comparable computational demands. Notably, across all the evaluated language transformers, DeiT persistently emerged as the most time-efficient model. In the configurations paired with RoBERTa, a pronounced increase in training durations was observed for all vision transformer variants relative to their training times with other language transformers. Particularly, the BEiT and RoBERTa combination required the longest training time, approximately 860 min. Overall, BEiT consistently demonstrated the longest training intervals, regardless of the language transformer it was combined with. This prolonged training period can be attributed to the inherent computational demands integral to BEiT's method of visual embedding integration.

Evaluation of Quantum Feature Maps in Multimodal Fusion

During the preliminary phase of our experiments, all model variants were uniformly integrated with the ZFM for comparative analysis. This quantum feature map was chosen due to its general applicability and compatibility across various model configurations. From this comparative evaluation, the BEiT + RoBERTa model variant emerged as the top-performing model, showcasing optimal results in both training efficiency and accuracy metrics. Furthermore, varying the number of qubits did not yield any significant changes in the model's accuracy.

Table 2 Performance metrics for BEiT + RoBERTa model with various feature maps

Model	Train Time (mins)	Val Acc	Test Acc
BEiT + RoBERTa (w/o quantum)	687.36	74.39	73.22
BEiT + RoBERTa (ZFM)	859.02	78.19	76.36
BEiT + RoBERTa (PFM)	1796.81	78.93	76.57
BEiT + RoBERTa (ZZFM)	924.36	79.42	77.51

In Table 2, we present the performance metrics of the BEiT + RoBERTa model variant when integrated with various quantum feature maps during the fusion phase. The study involved the utilization of three distinct feature maps: ZFM, PFM, and ZZFM. From the results, the BEiT + RoBERTa combination with the ZZ feature map achieved the highest validation and test accuracies of 79.52% and 77.61%, respectively. This represents an accuracy boost of approximately 1.15% from the baseline model using ZFM. This suggests that the ZZFM is particularly adept at capturing and representing the combined visual and linguistic features, thus enhancing the overall performance of the model on the VQA task.

The model variant without the quantum layer showed significantly lower validation and test accuracies of 74.39% and 73.22%, respectively. This highlights the substantial role of the quantum feature map in improving the model's performance. The PFM took nearly twice as long to train as the other two feature maps, with a duration of 1796 min. However, it did not exhibit any improvement in performance. Its test accuracy of 76.57% is only marginally higher than that of ZFM but remains below the results achieved with ZZFM. This points towards the computational intensiveness of PFM, which may not necessarily translate into a significant improvement in accuracy for this specific task. On the other hand, ZFM, with a training time of 859.02 min, achieved test accuracy comparable to PFM, indicating its efficiency in balancing computational demands with performance. Among the tested feature maps, the ZZ feature map demonstrated the best balance of training efficiency and performance, making it the most suitable choice for enhancing the BEiT + RoBERTa model variant in the context of the VQA task.

In our study, the quantum simulations were conducted in a noiseless environment, providing an idealized scenario for evaluating the model's performance. Under these conditions, the BEiT + RoBERTa model with the ZFM feature map achieved an accuracy of 76.36%. However, real-world quantum computations are subject to various types of noise, which can significantly impact the performance of quantum

algorithms. To provide a comprehensive analysis, we introduced depolarizing noise channels with a probability of 0.01 from the PennyLane library into our quantum simulations. Under these noisy conditions, the model achieved an accuracy of 74.96%. This decrease in accuracy highlights the importance of noise mitigation techniques and emphasizes the need for developing robust quantum algorithms that can perform well in practical, noisy environments.

Conclusion

The convergence of quantum computing with deep multimodal learning offers a promising approach for harnessing advanced computational capabilities in complex data integrations. In this work, we introduced a novel quantum-infused multimodal model which was further evaluated on the VQA task. The experimental results clearly highlight the efficacy of our proposed approach, showcasing promising advancements in integrating quantum computing with transformers. Our experiments show that the BEiT + RoBERTa architecture exhibits optimal performance, achieving an accuracy of 76.36% with the Z feature map, and an even more impressive accuracy of 77.61% with the ZZ feature map. These findings not only illustrate the potential of quantum interventions in enhancing vision language models but also highlight the role of quantum computing in the broader landscape of deep learning. However, it is crucial to acknowledge the current constraints posed by quantum computing hardware. Our model, despite its achievements, is still bound by these limitations. As quantum hardware continues to advance with ongoing developments, there is potential for scaling up the quantum components within our model, leading to further enhancements in performance. Future works could focus on incorporating quantum layers directly within the transformer blocks prior to fusion to further optimize the interaction and performance. Additionally, as quantum algorithms become more sophisticated, integrating them with state-of-the-art transformer architectures could unlock new areas of research. It might also be beneficial to explore the scalability of our model in larger datasets and other VL tasks beyond VQA. Additionally, as quantum hardware evolves, optimization techniques specifically tailored for quantum-infused models could be a promising direction to ensure efficient training and inference.

Data availability It is publicly available dataset.

Declarations

Ethical Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Conflict of interest The authors have no financial or non-financial conflicts of interest to disclose.

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